

access technologies (RAT), C-RAN and cloud self-organising network (C-SON) solutions.

As for networks, carriers will have to run optical fibre links at 20Gbps-40Gbps, or more, between remote antennae and server farms in some cases. For small areas such as auditoriums, 10Gbps Ethernet could be adequate. Solution-providers-proven advanced modem processing unit (MPU) solutions and modem programming language (MPL) for baseband and signal processing as a service (SPaaS) application, combined with Xilinx FPGAs and technologies present a fully-virtualised and programmable base station solution that addresses today's and future RAN requirements.

The virtual base station (vBS) solution is designed in accordance with ETSI NFV framework, which decouples and virtualises all software and hardware resources including base station L1/PHY software, x86 compute and hardware acceleration (HWA) platform to process real-time, signal processing type of workload. It enables realisation of end-to-end, fully-virtualised networks, from core to edge.

vBS will enable improved network performance, coverage and capacity, and could be deployed in numerous network configurations including C-RAN, traditional macro-cell sites, in-building/outdoor distributed antenna system deployments and even small-cell implementations.

The advent of FPGAs and solutions enables us to support our early customer engagements faster while offering maximum flexibility to address their needs. Emerging vBS requires high performance and programmability, while delivering low cost and power. The software stack runs on baseband unit servers together with an FPGA front-end processing board that emulates baseband signal processing flow.

The FPGA board integrates PCI Express, 10Gbps Ethernet, common public radio interface, optical connections and so on, which enables it to bridge telecom and IT datacenter domains. As an accelerator, the FPGA implements many key algorithm units required by baseband signal processing, which increases the system computing power greatly.

RapidIO protocol

Each radio node in the architecture serves a set of users in a small- or macro-cell configuration. In both traditional and distributed architectures, RapidIO protocol con-

RapidIO in this case is expected to provide best-in-class performance.

Two major flows in the access network include transmission from mobile to base station (up-link flow) and reception at mobile from the base station (downlink flow). Retransmission, in the form of hybrid-automatic-repeat-request (HARQ), might be required in case of transmission errors as part of LTE/LTE-A protocol.

HARQ timing of around 4ms is actually much lower than LTE/LTE-A round-trip latency. If the interconnect fabric between processing nodes supports superior flow control and fault-tolerance capability, it is further possible to minimise the number of HARQ retransmissions, resulting into lower latency. This provides a differentiation point for OEMs and eventually leads to better quality of experience for the end user.

To support handover, load balancing and interference management, above major flows further include exchange of information related to handover, channel quality and load indication between various processing units at the base station.

Exchange of information between baseband units should be supported with lowest deterministic latency and guaranteed delivery. This allows demand on the network and interference between various users to be identified without error and in a timely manner. This also enables low-latency handover with reliability as users cross cell boundaries.

To meet the requirements in C-RAN and a small cell, in particular, reliable load management, hand-over and interference management, OEMs are evolving base station designs by taking advantage of RapidIO's interconnect features and advancements in systems on chip (SoC), memory and radio sub-system components.

For the interconnect, there are two important functions to consider:

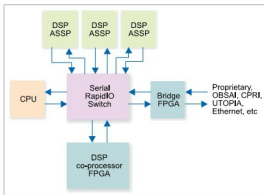


Fig. 2: RapidIO protocol switch interface

nects multiple processing units (for example, DSPs, SOCs, ASICs and FPGAs) in channel or baseband cards. The protocol ensures guaranteed delivery with lowest latency of around 100nsec between any two processing nodes.

In a typical RAN architecture, once the signal processing related to a particular radio interface is completed, data is transported between the baseband and the radio using CPRI protocol. In a distributed architecture, CPRI protocol may not lead to a cost-effective implementation of load management and interference control, since, by definition, CPRI protocol does not provide a standardised low-latency packet based switching capability that can be used to distribute traffic across multiple baseband cards from radios.